

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID - 26222**
- **Problem Statement Title -** Student Innovation-Submit your ideas to address the growing pressures on the city's resources, transport networks, and logistic infrastructure.
- **Theme -** Transportation
- **PS Category-** Hardware
- **Team ID - 181666**
- **Team Name - SmartRider**

Problem

Existing bus tracking systems mainly provide **bus location and arrival information**, but do not show the *current crowd level* inside the bus.

As a result, passengers may wait for an overcrowded bus, even when another bus with more available space is approaching.

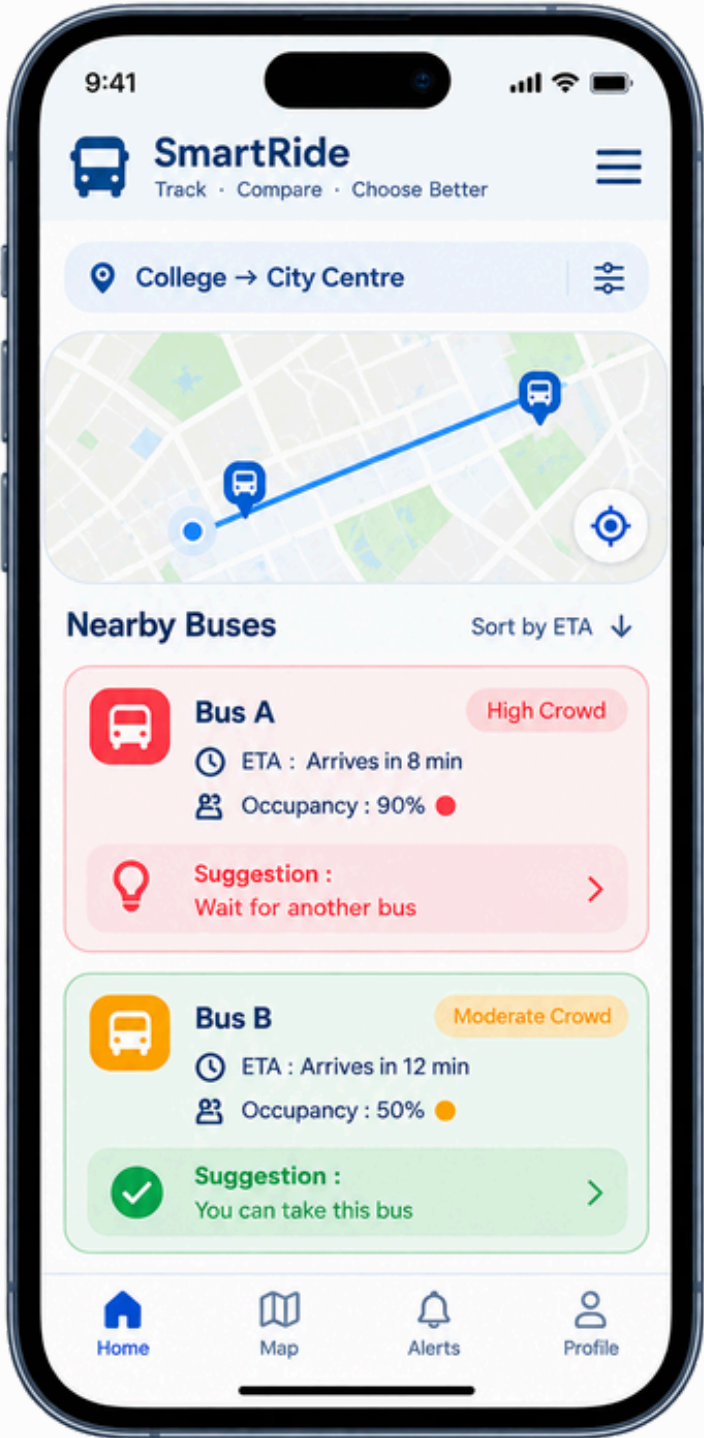
Solution

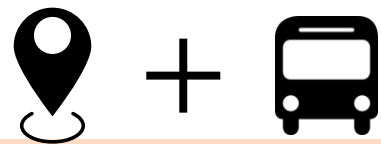
Introducing **SmartRide** __ a real-time bus tracking system designed to make public transport more convenient. Unlike conventional systems that mainly provide bus location and ETA, SmartRide additionally provides **real-time occupancy levels and actionable suggestions**

■ Difference - How it varies from existing model

Existing approach	Limitation	SmartRide improvement
Bus location + ETA	No crowd information	Location + ETA + occupancy
Multiple buses on route	Difficult to compare	Bus-to-bus comparison
Passenger decides manually	Limited information	Actionable travel suggestion

Track → know the crowd → choose better





Data input

(Sensing and collection)

Captures real-time bus location, passenger occupancy, and route information.

Technology Used:

GPS+Load Sensor(major component)+Ardiuno



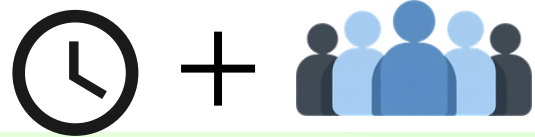
Data processing

(validate and filter)

Validate incoming data, filters noise and cleans the raw data for further analysis.

Technology Used:

Embedded controller+ data processing



Estimation

(Analysis)

Estimates crowd and arrival time, delays using sensor and GPS data with future ML based prediction.

Technology Used:

python+database
(current: sensor based estimation)



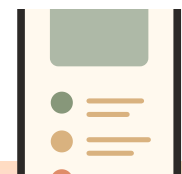
Recommendation

(Decision making)

Compares buses based on the ETA+occupancy+delay to recommend the most suitable bus.

Technology Used:

Alogorithm/ decision based logic



User Output

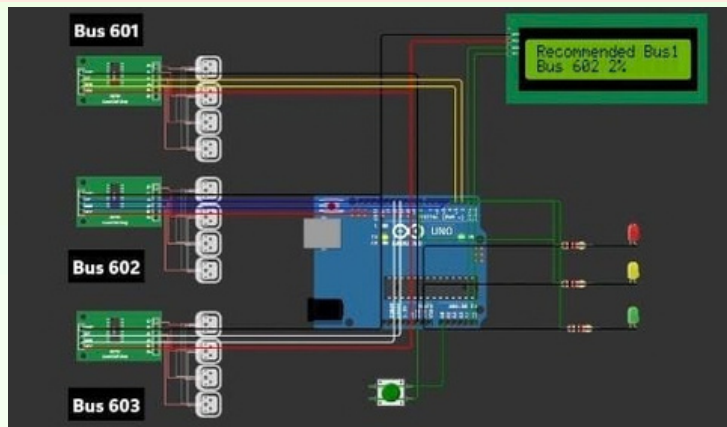
(Presentation)

Presents the real-time information and provides the alternative buses.

Technology Used:

LCD(prototype), web/mobile app (future)

Current Implementation



Implemented and tested a working prototype with load cell integration validated through wokwi

working prototype model



Future Implemenatation

Develop a web/mobile app user interface for real-time tracking and smart bus recommendation with the help of GPS integration and historical data.

Backend can be improved by integrating AI-ML prediction in the future

FEASIBILITY AND VIABILITY

- **Technical Feasibility**

We can build the system using a microcontroller, sensors, switches, and control circuits

- **Operational Feasibility**

The device can work automatically, so less manual control is needed.

- **Resource Feasibility**

The device can work automatically, so less manual control is needed.

- **Integration Feasibility**

The device can work automatically, so less manual control is needed.

- **Cost viability**

It can reduce operating and maintenance costs

- **Performance viability**

It provides faster, accurate, and reliable switching

- **Energy viability**

It can reduce power loss and save energy.

- **Future viability**

In the future, it can be improved using IoT, AI, remote monitoring, and predictive maintenance

IMPACT AND BENEFITS

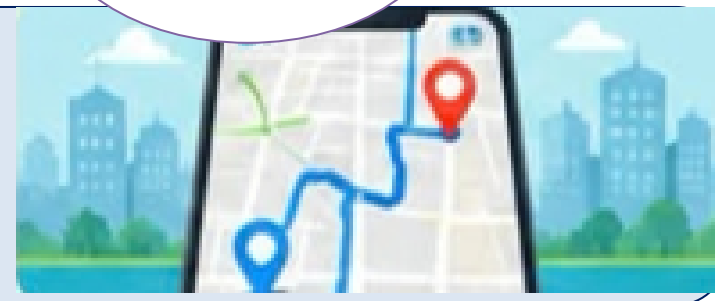


Impacts

Reduces Traffic Jams
and saves travel time ,energy and
Manual effort and safety.



Helps People
Find better and faster routes &
Solve manual , slow & inefficient
Switching problems



Reduces
Fuel usage and transportation
costs, reduce operational &
maintenance cost



**Makes city
transportation**
And deliver services more
efficient & improve system efficiency



Benefits

**Reduces air
pollution**

And vehicle emissions reduces
power loss



Makes daily travel

Safer and easier for people & it
provides faster switching , lower
power loss



Helps cities use

Their roads and resources
better



Supports

The development of a smart and
Sustainable city and industries,
business & electrical systems users



Measurable Impact

- **Occupancy Awareness**
- **Route Decision Time**
- **Reduced Unnecessary Travel**
- **System Accuracy**

Bongiovi, G., Galvão Dias, T., Nauri Junior, J., & Campos Ferreira, M. (2026).

“Passenger occupancy detection and estimation in public transport: a systematic review of technologies and predictive models.”

Ribeiro, M., et al. (2023).

Wireless Sensor Network Architecture for Passenger Counting in Public Transportation.

Murdan, A., et al. (2020).

Low-cost bus seating information technology system.